14. Fabricating Systems A production plant has two fabricating systems, both of which are maintained at 2-week intervals. However, one system is twice as old as the other. The number of finished products fabricated daily by each of the systems is recorded for 30 working days, with the results given in the table. Do these data present sufficient evidence to conclude that the variability in daily production warrants increased maintenance of the older fabricating system? Use the *p*-value approach.

New System	Old System
$\bar{x}_1 = 246$	$\bar{x}_2 = 240$
$s_1 = 15.6$	$s_2 = 28.2$

① $H_0: \sigma_1^2 = \sigma_2^2 \quad H_a: \sigma_1^2 \neq \sigma_2^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{S_2^2}{S_1^2} \sim F_{(n_2-1),(n_1-1)}$

④ $F^* = \frac{28.2^2}{15.6^2} \approx 3.2678$

⑤ $F_{29,30;0.05} = 2.66 < F^*$

p-value < 0.05

⑥ reject H_0.

16. SAT Scores The SAT subject tests in chemistry and physics for two groups of 15 students each electing to take these tests are given here.

Chemistry	Physics
$\bar{x} = 784$	$\bar{x} = 758$
$s = 114$	$s = 103$
$n = 15$	$n = 15$

To use the two-sample *t*-test with a pooled estimate of σ^2, you must assume that the two population variances are equal. Test this assumption using the *F*-test for equality of variances. What is the approximate *p*-value for the test?

① $H_0: \sigma_1^2 = \sigma_2^2 \quad H_a: \sigma_1^2 \neq \sigma_2^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{S_1^2}{S_2^2} \sim F_{(n_1-1)(n_2-1)}$

④ $F^* = \frac{114^2}{103^2} = 1.22 \quad df_1 = 14 = df_2$

⑤ $F_{15,14;0.1} = 2.01 > 1.22 \quad$ p-value > 0.1

⑥ p-value $> 0.1 > 0.05 \Rightarrow$ non-reject H_0

22. Stock Risks The closing prices of two common stocks were recorded for a period of 15 days. The means and variances are

$\bar{x}_1 = 40.33 \qquad \bar{x}_2 = 42.54$

$s_1^2 = 1.54 \qquad s_2^2 = 2.96$

a. Do these data present sufficient evidence to indicate a difference between the variabilities of the closing prices of the two stocks for the populations associated with the two samples? Give the *p*-value for the test and interpret its value.

b. Construct a 99% confidence interval for the ratio of the two population variances.

a.

① $H_0: \sigma_1^2 = \sigma_2^2 \quad H_a: \sigma_1^2 \neq \sigma_2^2$

② $\alpha = 0.05$

③ $F_{STAT} = \frac{S_2^2}{S_1^2} \sim F_{(n_2-1),(n_1-1)}$

④ $F^* = \frac{2.96}{1.54} = 1.92 \quad df_1 = df_2 = 14$

⑤ $F_{15,14;0.1} = 2.01 > 1.92 \quad$ p-value > 0.2

⑥ p-value $> 0.2 > 0.05 \Rightarrow$ non-reject H_0

b.

$$\left[1.92 \times \frac{1}{F_{15,14;0.01/2}},\ 1.92 \times F_{15,14;0.01/2}\right]$$

$$= \left[1.92 \times \frac{1}{4.25},\ 1.92 \times 4.25\right) = [0.45,\ 8.16]$$